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Shortest Cable Routing of Heliostats in Solar Tower Power Plants

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Aachen, im September 2018

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1 Introduction

In this work we present two heuristics that compute valid solutions to an optimization problem regarding a cabling problem in solar tower power plants. Such a plant focuses sunlight with the help of heliostats onto a receiver that is installed at the top of the solar tower. This energy heats up a medium like oil or sodium in order to power a turbine which generates electric energy in return. Such a plant consists of a solar tower and heliostats that have to be connected to the solar tower with two types of cables, namely a power and data cable. Both cables are needed at each heliostat as the sun's position has to be taken into account for an efficient exploitation of solar energy. Regarding the cabling we have several limitations with respect to connectivity and planarity constraints. First, cables have restrictions regarding their maximum lengths between heliostats. Moreover, there are capacity constraints for each cable, namely that cables can only connect so many heliostats. Second, the planarity constraint is enforced in order to have a low maintenance cost when digging up trenches.

Another source of energy is utilized in offshore wind farms for which the cabling problem has been studied in [5, 2, 4]. The same constraints like planarity are also present in their cabling problem. Our work builds up on two bachelor's theses that studied the solar tower power plants [6, 10].

In the following we will give a formal definition of our problem and present two heuristics that apply the local search optimization method [1, 11]. This approach was chosen due to the high complexity of the cabling problem, which is especially difficult to solve for large instances that inhibit a lot of heliostats. Then, an evaluation part is followed where we have applied both heuristics onto the PS10 solar tower power plant [7, 8]. Afterwards we give a short summary and possible future work.

2 Problem statement

Formally, we are given a complete, directed graph $G = (V, A)$ where $V = \{T\} \cup V_H$ with T being our root solar tower and $V_H := \{v_1, \ldots, v_n\}$ denotes the set of heliostats. Note that we work in a metric space $(\mathbb{R}^2, d)$, thus we denote with $v_x \in \mathbb{R}$ and $v_y \in \mathbb{R}$ both coordinates for each vertex $v \in V$. Moreover, the metric is simply the euclidean distance

$$d(v, w) := \sqrt{(v_x - w_x)^2 + (v_y - w_y)^2}$$

between nodes $v, w \in V$. In other words, each vertex $v \in V$ has a specified position in a 2D plane.

Before a cable can be installed a trench has to be constructed by workers for which we have to pay installation costs. Let us denote these costs per meter with $c^{\text{trench}} \in \mathbb{R}^+$. In other words, the total costs of providing a trench between $v, w \in V$ is
$c^{\text{trench}}_{(vw)} := d(v, w) \cdot c^{\text{trench}}$.

2.1 Data cable and local control units

We can install different types of cables which have length and connection restrictions depending on its type. For this matter let D be the set of data cables with costs $c_d^{\text{data}} \in \mathbb{R}^+$ per meter for each cable $d \in D$. We further define for readability the costs $c_{d,(vw)}^{\text{data}} := d(v,w) \cdot c_d^{\text{data}}$ of installing a cable of type d on the edge $(vw) \in A$. Furthermore, each data cable $d \in D$ has a length restriction $L_d^{\text{data}} \in \mathbb{R}^+$ and a restriction on how many heliostats it can connect to, that is, $M_d^{\text{data}} \in \mathbb{N}$ for $d \in D$. We denote this property by *capacity restriction*.

Aside from the cables we have to build *local control units* (LCU) at each heliostat. These LCUs are used for communication processing and may allow branching by e.g. installing an 8- or 16-port LCU at an heliostat. Each of these LCU has connectivity-restrictions for each cable type $d \in D$. For this matter, let S be the set of all LCUs where each $s \in S$ has installation costs $c_s^{\text{lcu}} \in \mathbb{R}^+$. The connectivity or vertex-degree restrictions are given by $l_{s,d}^-, u_{s,d}^- \in \mathbb{N}$ and $l_{s,d}^+, u_{s,d}^+ \in \mathbb{N}$ for each LCU $s \in S$ and cable type $d \in D$, where the superscript $-$ denotes in-going and $+$ out-going cables, respectively. Both type of restrictions state that the LCU $s \in S$ has to connect at least $l_{s,d}^-$ in-going cables of type $d \in D$, whereas $u_{s,d}^+$ is the upper bound of outgoing cables of type $d \in D$ at an heliostat that installed LCU $s \in S$.

2.2 Power cables

Like the data cables, each power cable has a length and capacity restriction. Let P be the set of power cables with costs $c_p^{\text{power}} \in \mathbb{R}^+$ per meter, length restrictions $L_p^{power} \in \mathbb{R}^+$ and capacity restrictions $M_p^{power} \in \mathbb{N}$ for $p \in P$. Furthermore, we define the costs $c_{p,\{vw\}}^{\text{power}} := d(v,w) \cdot c_p^{\text{power}}$ that are the required costs of installing a power cable $p \in P$ along the edge $\{vw\} \in E$.

In contrast to the data cables we do not to consider any type of switches for the power cables.

2.3 Cabling problem

For a directed tree $D(T) = (V, A)$ with root $T \in V$ we denote the unique path from the root T and a vertex $v \in V$ by $path(T, v)$. Moreover, we define the following sets for $v \in V$:

- $ancestors(v) := V(path(T, v)) \setminus \{v\}$
- $descendants(v) := \{w \in V | \exists path(v, w) \text{ in } G\}$

Problem 1 (Cabling Problem) *Given a complete, directed graph $G = (V, A)$, a set of data cables D and power cables P. Two directed, rooted trees $B_{data}(T), B_{power}(T) \subseteq G$ with cable assignments $assign_{data} : A(B_{data}) \to D$ and $assign_{power} : A(B_{power}) \to P$ as well as the LCU assignment $assign_{lcu} : V(B_{data}) \to S$, where the root is the solar tower $T \in V$ for both trees, are to be constructed such that the following properties are fulfilled:*

Data cable $\forall a =: (uv) \in A(B_{data})$: *let* $d_a := assign_{data}(a)$

- $|descendants(v)| + 1 \leq maxCapacity(d_a)$
- $d(u, v) \leq maxLength(d_a)$

LCU $\forall h \in V_H$: *let* $lcu_h := assign_{lcu}(h)$

- $\forall d \in D$: $l^-_{lcu_h,d} \leq \sum_{a \in \delta^-(h): assign_{data}(a)=d} 1 \leq u^-_{lcu_h,d}$
- $\forall d \in D$: $l^+_{lcu_h,d} \leq \sum_{a \in \delta^+(h): assign_{data}(a)=d} 1 \leq u^+_{lcu_h,d}$

Power cable *for all* $a' := (u, v) \in A(B_{power})$:

- $|descendants(v)| + 1 \leq maxCapacity(assign_{data}(a'))$
- $d(u, v) \leq maxLength(a')$

Intersection $\nexists e, f \in A(B_{data}) \cup A(B_{power})$ *that intersect with each other.*

with the objective of minimizing the total cost:

$$\min \left\{ \sum_{(vw) \in A(B_{data}) \cup A(B_{power})} c^{trench}_{(vw)} + \sum_{(vw) \in A(B_{data})} c^{data}_{d,(vw)} + \sum_{(vw) \in A(B_{power})} c^{power}_{p,\{vw\}} \right\}$$

3 Data

We present real world data regarding costs per meter for various data as well as power cables. Moreover, many different LCUs and trench/installation costs are provided for which we can test our heuristics on. The experiments are running on the PS10 solar tpower power plant instance [7, 8] for which the coordinates are provided, a sketch of this instance can be seen in fig. 1.

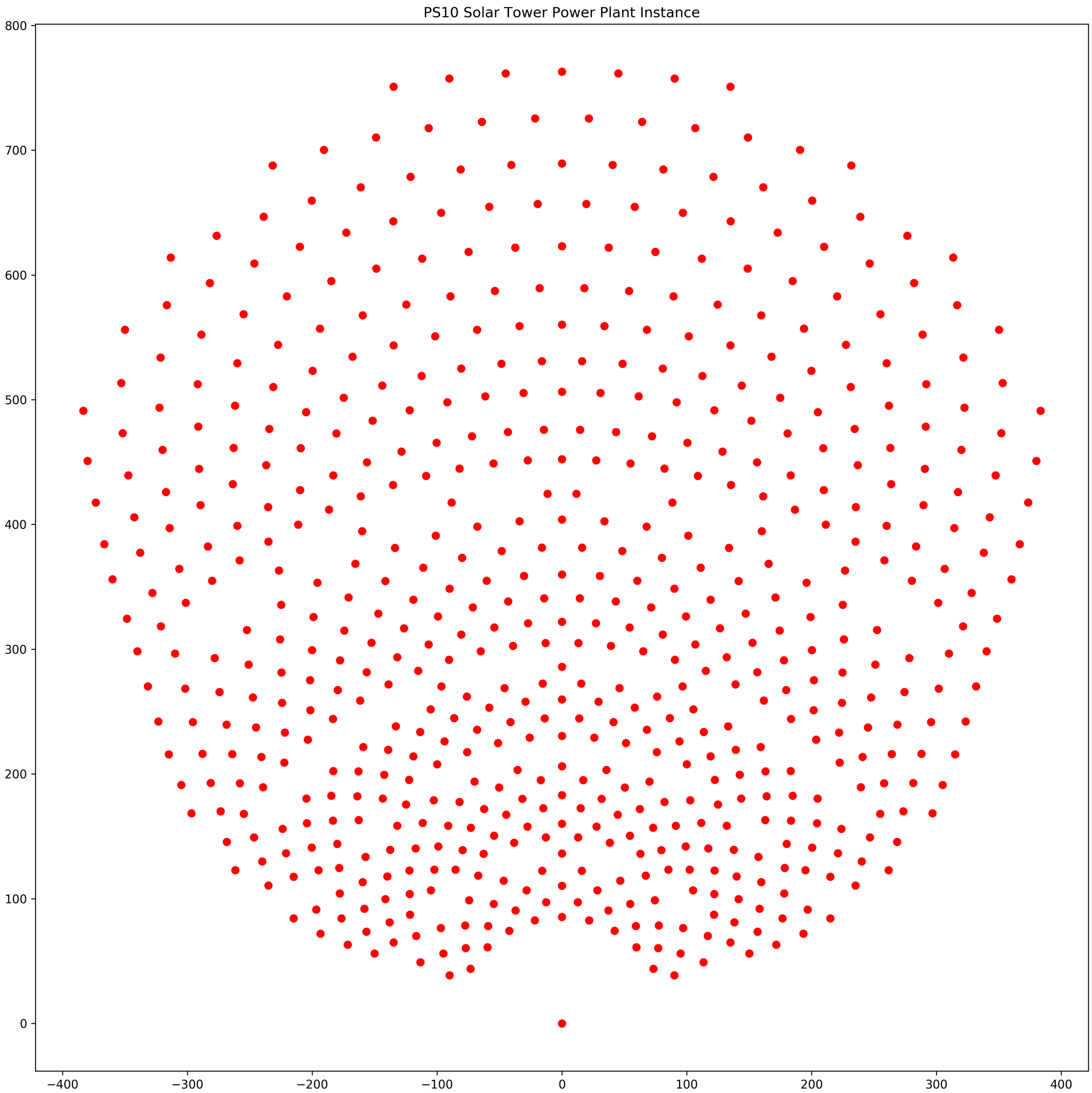

Figure 1: PS10 instance with its heliostats and the solar tower at coordinate $(0, 0)$.

3.1 Trench costs

For the trench costs we are considering four different costs, each depending on the country we are located in. Each trench has to be constructed by workers and has a depth of 1m as well as 1m in width and length.

Table 1: Trench costs for different countries

Country	Costs [€/m]
Australia	50
Spain	25
South Africa	10
United Arab Emirates	10

3.2 Data Cable

Regarding the data cable we are able to install either a copper or fiberglass cable. The copper cable is much cheaper however it has harsh restrictions regarding its usage, making it only usable for endpoint connections. The fiberglass on the other hand can connect an arbitrary amount of heliostats and has no length restrictions.

Table 2: Possible data cables with their cost and restrictions.

Data cable type	Max. capacity	Max. length [m]	Price [€/m]
Copper	2	100	0.7
Fiberglass	∞	∞	2

3.2.1 Local Control Units

Exactly one local control units is installed at every heliostat so that communication is enabled. All LCUs with the exception of an endpoint has exactly one ingoing fiberglass cable. The endpoint has exactly one ingoing copper cable.

Table 3: Local control units with their degree and cable restrictions for both cable types and direction.

LCU type s	Price [€/piece]	Data cable d			
		Copper		Fiberglass	
		$[l^-_{s,d}, u^-_{s,d}]$	$[l^+_{s,d}, u^+_{s,d}]$	$[l^-_{s,d}, u^-_{s,d}]$	$[l^+_{s,d}, u^+_{s,d}]$
Endpoint	10	$[1, 1]$	$[0, 0]$	$[0, 0]$	$[0, 0]$
Conductor	100	$[0, 0]$	$[0, 0]$	$[1, 1]$	$[0, 1]$
8-port copper	800	$[0, 0]$	$[0, 7]$	$[1, 1]$	$[0, 1]$
8-port fiberglass	800	$[0, 0]$	$[0, 0]$	$[1, 1]$	$[0, 8]$
16-port copper	1500	$[0, 0]$	$[0, 15]$	$[1, 1]$	$[0, 1]$
16-port fiberglass	1500	$[0, 0]$	$[0, 0]$	$[1, 1]$	$[0, 16]$

Therefore, an 8-port copper LCU for example has up to 7 'outgoing' copper connections and up to one 'outgoing' fiberglass one.

3.3 Power Cable

The power cables are not restricted with LCUs. However, maximum capacity constraints have to be taken into account — as opposed to glasfiber cables that do not have any capacity restrictions. Hence, we have to use more than one power cable for instances with more than 250 heliostats. Therefore we have to partition the heliostats field into partitions that is non-trivial. A simple heuristic is presented in section 4.1.

Table 4: Power cables with their connectivity limitations and prices.

Power cable type	Max. capacity	Max. length [m]	Price [€/m]
NYY-J 3x2.5 RE	56	38.36	0.58
NYY-J 3x4 RE	73	47.08	0.87
NYY-J 3x6 RE	92	56.04	1.24
NYY-J 3x10 RE	124	69.30	1.95
NYY-J 3x16 RE	162	84.87	3.13
NYY-J 3x25 RM	209	102.79	5.19
NYY-J 3x35 RM	250	120.31	6.90

4 Heuristics

We only consider fiberglass cables for the data connections for simplicity due to their nature of being unrestricted in both length and capacity. Moreover, we compute solutions for which both data and power cable are installed

In the following both sections we present our heuristics that are based on the local-search optimization method [1, 11]. The basic structure of such an algorithm is shown in algorithm 1. Basically we first compute an initial solution from the input and afterwards improve it stepwise until some stopping criterion is met. In principle the stepwise improvements are performed locally to *sol* such that each step yields a valid solution to our optimization problem.

Algorithm 1 A general framework of a local search heuristic.

```
procedure LOCAL-SEARCH(input)
    init := COMPUTE-INITIAL-SOLUTION(input)
    sol := init
    while stopping criterion not met do
        x := COMPUTE-NEXT-NEIGHBOR-SOLUTION(sol)
        if c(x) < c(sol) then
            sol := x
        end if
    end while
    return sol
end procedure
```

One may interpret this by jumping from one point in the valid solution search space to another one in a certain local neighborhood. This interpretation is depicted in fig. 2 where s_0 is the initial solution that we improve in a stepwise-manner by selecting a neighboring solution s_1 in its neighborhood $N(s_0)$. Obviously s_1 has lower total costs than s_2, however, depending on the neighborhood size $|N(s_0)|$ it may be a computationally demanding task. Therefore, while generating the neighboring solutions in $N(s_0)$ we terminate the search process as soon as we encounter a solution $s_1 \in N(s_0)$ with $c(s_1) < c(s_0)$. This search strategy is called *first-improvement* as we do not generate the best neighboring solution but are satisfied with any kind of improvement. This process is then continued until some stopping criterion is met. In particular, one may terminate after $i \in \mathbb{N}$ steps or in our case terminate as soon as $N(s_i)$ does not yield any improving solution, hence s_i is called *locally optimal*.

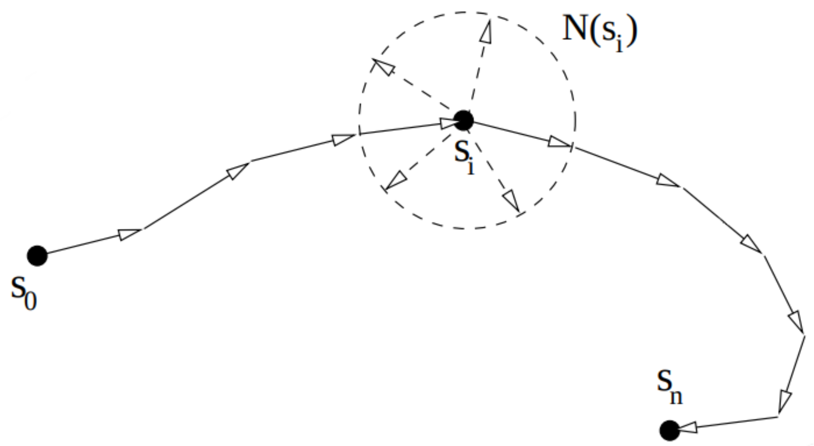

Figure 2: The search space where each point describes a valid solution to our optimization problem. We start from the initial solution s_0 improve it stepwise. For this matter we consider 'neighboring' solutions in $N(s_i)$ at the current solution s_i. When no neighboring solution is improving (i.e. has less total costs than current solution) we terminate our algorithm as s_n is a locally optimal solution.

4.1 Partitioning

Because the largest power cable has a capacity of 250 heliostats and instances like the PS10 have more heliostats we have to use additional power cables in order to connect all heliostats, hence a partioning of the heliosats has to be considered in addition. We perform a simple heuristic for this problem by computing the angles α_h between the reference vector $(1, 0)^T$ and a heliostat $h \in V_H$ like in fig. 3. Afterwards, the partioning is the result of sorting the angles and splitting them according to a parameter $K \in \mathbb{N}$. All in all, this procedure has a time complexity of $\mathcal{O}\left(|V_H|^2 \log\left(|V_H|\right)\right)$.

Important to note is that we can compute initial solutions for the local search procedures independently on each partition, therefore obtaining K initial solutions for K

partitions.

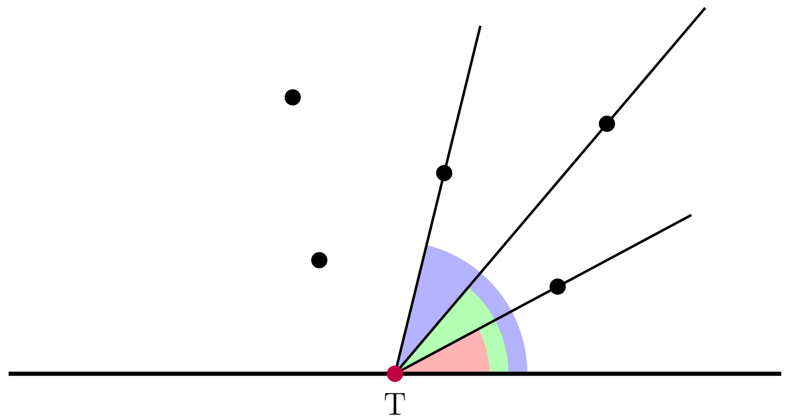

Figure 3: Sort heliostats by computed angles. For $K \in \mathbb{N}$ partitions we obtain $\left\lceil \frac{|V_H|}{K} \right\rceil$ heliostats in each partition.

Because this heuristic is a hard assignment of the heliostats to their respective partition we have to consider exchange-operations in the local search algorithms so that non-convex partitions are able to arise. Otherwise we limit ourselves to the hard-assignments and local search solutions are poor with regards to their quality.
More formally we compute *partitions* $G_k := (V_k, E_k)$ with

- $V_k := \{T\} \cup V_{H,k}$
- $E_k := \{\{vw\} | v, w \in V_k\}$

where $V_{H,k} \subseteq V_H$ is the set of heliostats belonging to partition $k \in \{1, \dots, K\}$ that is the result of sorting the angles. Note that $V_{H,i} \cap V_{H,j} = \emptyset$ for $i \neq j$, $1 \leq i, j \leq K$ and $\bigcup_{k=1}^{K} V_{H,k} = V_H$.

4.2 Greedy Power Cable Assignment

4.3 Minimum Spanning Tree Approach

The fact that each heliostat has to be connected to the solar tower T with minimal costs infers utilizing a minimum spanning tree approach. We just need to consider the additional constraints of planarity as well as connectivity constraints. We thus choose Prim's algorithm [9, 3] to compute an initial, valid solution to our problem. Afterwards we compute a greedy power cable assignment which yields a valid solution to problem 1, like descriped

For the local neighborhood improvement steps we choose the *cycle-neighborhood*

$$N(B) := \{\tilde{B} \in \mathcal{B} \mid |E(\tilde{B}) \setminus E(B)| = 1\}$$

for a spanning tree $B \subset G$ and $\mathcal{B}$ denotes the set of all spanning trees in G.

For an update step we thus delete an edge $e \in E(B)$ in our current tree B and reconnect the disconnected partition by a new edge $f \in E \setminus (E(B) \cup \{e\})$ which yields $\tilde{B}$. By doing this we effectively move from B to another spanning tree $\tilde{B}$ in the search space. In order to improve our current solution we always choose an edge f with $c(f) < c(e)$. If no such edge exists, we can terminate the local search procedure, hence B is a locally optimal solution. Such an update step is sketched in fig. 4.

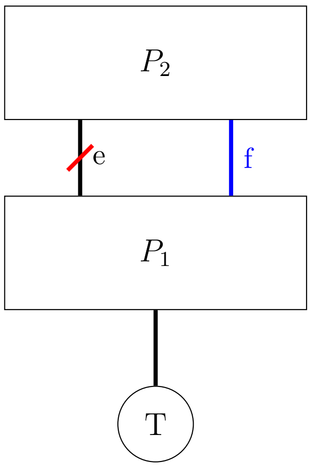

Figure 4: Consider the cycle neighborhood $N(B)$ for some spanning tree $B \subset G$, that is, delete edge e and reconnect the resulting partitions P_1 and P_2 by adding an edge f with $c(f) < c(e)$, yielding a spanning tree with lower costs.

Moreover, due to changes in capacities we also need to consider the power cabling in the local search improvement step. Let us take up the case of deleting the edge $(u, w) \in E(B)$ in our current spanning tree $B \in \mathcal{B}$. Moreover, the edge $(v, w) \notin E(B)$ is considered as replacement. This procedure is sketched in fig. 5.

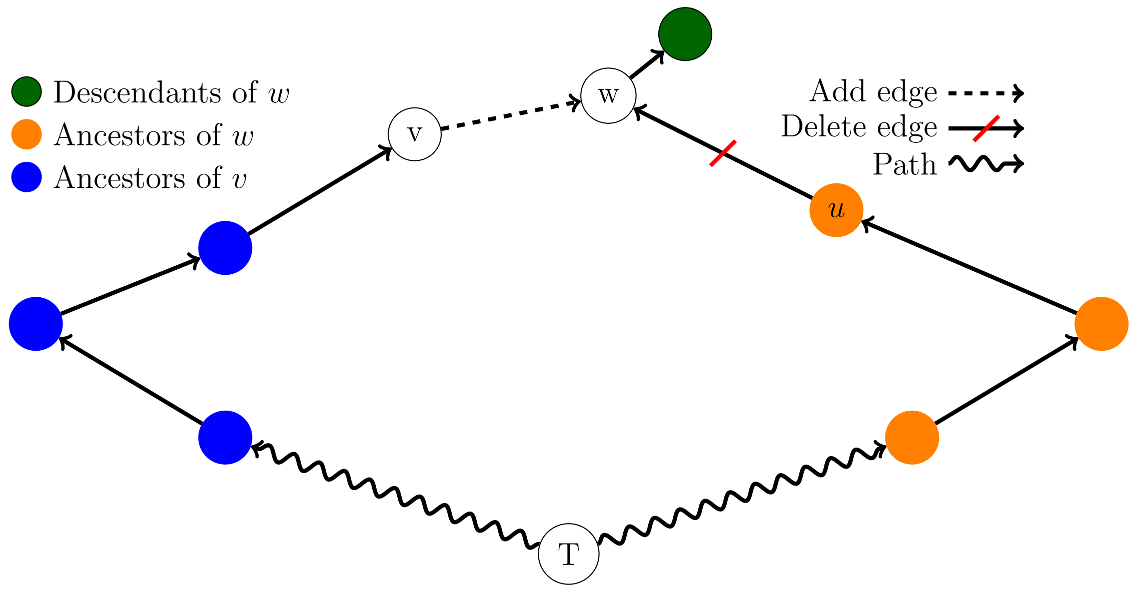

Figure 5: For power cables we need to consider up- and downgrades in each improvement step, consequently potential savings by switching to lower-quality cables are made.

Each power cable on $path(T, v)$ has to be checked for a downgrade, on the other hand it may be necessary to upgrade power cables on $path(T, u)$ in order to cope with the additional capacity of $w \in V_H$ and its descendants.

4.3.1 Downgrade

For each edge $(r, s) \in path(T, u)$ we check whether we can switch from $assign_{\text{power}}((r, s)) \in P$ to a cheaper power cable $p' \in P$ such that

- $|descendants(s)| + 1 \leq maxCapacity(p')$
- $d(r, s) \leq maxLength(p')$

4.3.2 Upgrade

If the additional capacity cannot be satisfied by the current power cable $assign_{\text{power}}((r, s))$ for an edge $(r, s) \in path(T, v)$ we need to upgrade to a higher-capacity cable $p' \in P$ such that

- $|descendants(s)| + 1 \leq maxCapacity(p')$
- $d(r, s) \leq maxLength(p')$

Therefore, we need to compute these additional cost differences of re-installing power cables during the local search improvement step.

If these capacity or length constraints cannot be satisfied for any power cable in down- or upgrades that are necessary to have a valid solution, then the local search move of adding (v, w) and deleting (u, w) is not feasible and thus omitted.

4.4 Hamiltonian Path Approach

If every heliostat is incident to maximal two cable connections we are able to use conductors only, which reduces the amount of LCU costs we are installing (see table 2). Therefore we consider a Hamiltonian path as a basis for the second heuristic.

4.4.1 Initial Solution

The initial solution is built greedily by starting from the solar tower T and extending the hamiltonian path step by step. For this matter we consider a *partial path* p of vertices of length $i \in \mathbb{N}$, that is, $p = (v_0, v_1, v_2, \ldots, v_i)$, for heliostats $v_j \in V_H, 1 \leq j \leq i \leq |V_H|$, $v_0 := T$. For arbitrary partial paths we define the distance as the sum of euclidean distances, that is

$$Dist\left((v_0, v_1, v_2, \ldots, v_i)\right) := \sum_{k=0}^{i-1} d(v_k, v_{k+1})$$

Given a partial path $p = (v_0, v_1, \ldots, v_i)$, $0 \leq i \leq |V_H|$ we additionally introduce the *extended path* for an unvisited heliostat $v \in V_H \setminus V(p)$ that is placed after the m-th vertex in the partial path p:

$$ExtPath(p, v, m) := (v_0, v_1, \ldots, v_m, v, v_{m+1}, \ldots, v_i)$$

for $1 \leq m \leq i$. In other words we may append a heliostat at the end of p or insert it into the path directly, effectively adding edges $\{v_{m-1}, v\}$, $\{v, v_{m+1}\}$ and deleting the existing edge $\{v_{m-1}, v_{m+1}\}$.

Therefore, given a partial path $p = (v_0, v_1, \ldots, v_i)$ we can extend it by computing the minimal distance of all possible extended paths:

$$\forall m \in \{1, \ldots, i\}\ \forall v \in V_H \setminus V(p)\ :\ \min\left\{Dist(ExtPath(p, v, m))\right\}$$

This search procedure can be computed in $\mathcal{O}(|V_H|^2)$ time in each step until all heliostat are visited. We also simply add a constant of 100 to the edge weights, thus effectively taking the costs of conductors into account. As opposed to the initial minimum spanning tree computation in section 4.3 we do not need to explicitly check for intersections during the computation because the partitions form a convex polygon and the edge weights are still metric.

For a partition $G_k \subset G$ we can compute an initial hamiltonian path by using algorithm 2.

Algorithm 2 Greedy construction of a hamiltonian path for a partition k.

procedure Init-Hamilton($G_k = (V_k, E_k)$)
    $p := (T)$
    while $|p| < |V_{H,k}|$ **do**
        $M[v, m] := \infty \qquad m = 1, \ldots, i, \quad v \in V_{H,k} \setminus V(p)$ ▷ init. value matrix
        for $m = 1, \ldots, i$ **do**
            for $v \in V_{H,k} \setminus V(p)$ **do**
                $M[v, m] := Dist(ExtPath(p, v, m))$
            end for
        end for
        $v^*, m^* := \arg\min_{v,m} \{M[v, m]\}$
        $p := ExtPath(p, v^*, m^*)$
    end while
end procedure

4.4.2 Local Neighborhood Search

For each partition $k = 1, \ldots, K$ we obtain a hamiltonian path $H_k \subset G_k$. Now, consider moving a vertex from a hamiltonian path H_i to a path H_j for $1 \leq i, j \leq K$. In particular consider the graph construct in fig. 6 where $v, w \in V(H_i)$ and $x, y, z \in V(H_j)$.

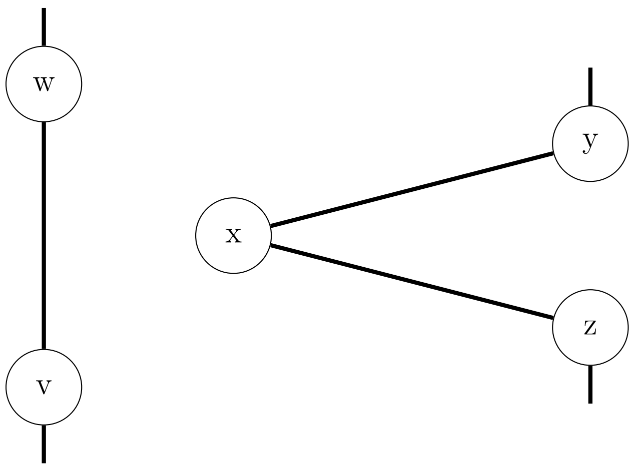

Figure 6: Starting point of the local search step. We denote the left hamiltonian path that contains the vertices v and w as H_i, the right one with H_j.

In order to move the vertex $x \in V(H_j)$ to the hamiltonian path H_i we have to delete the edges $\{x, y\}, \{x, z\} \in E(H_j)$ to isolate the vertex x as well as $\{v, w\} \in E(H_i)$. If x is a leaf then we only have to remove one edge in H_j.

Afterwards, we connect x to the hamiltonian path H_i by constructing the edges $\{v, x\}$ and $\{w, x\}$. Further, we might have to add the edge $\{y, z\}$ to $E(H_j)$ if x was not a leaf, therefore having two neighbors $y, z \in V(H_j)$. This procedure is sketched in fig. 7.

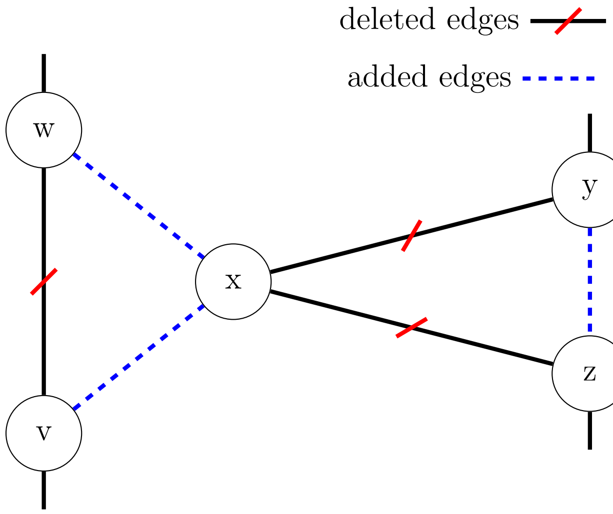

Figure 7: Moving $x \in V(H_j)$ requires deletion and creating edges in both hamiltonian paths H_i and H_j.

This improvement step is checked for all configurations of vertices $v, w \in V(H_i)$ and $x, y, z \in V(H_j)$ where z might be non-existent. In particular, we check for every configuration the cost difference

$$\begin{aligned} R := & c(x, w) + c(x, v) + c(y, z) \\ & - c(v, w) - c(x, y) - c(x, z) \end{aligned}$$

In the case of z being non-existent we leave out the terms $c(y, z)$ and $c(x, z)$. We further introduce a threshold $Q \in \mathbb{Q}^-$ and perform the local search in a first-improvement fashion. That is, if we encounter a configuration with $R \leq Q$ then we update our solution by moving x to the respective hamitonian path. This step is repeated until no configuration exists for which the threshold Q is not fullfilled, therefore having a locally optimal solution. The threshold R is introduced to accelerate termination of the algorithm and is set $Q := -1.0$ in our work, in other words we only perform the improvement if a newly found configuration is at least 1€ better in terms of cost.

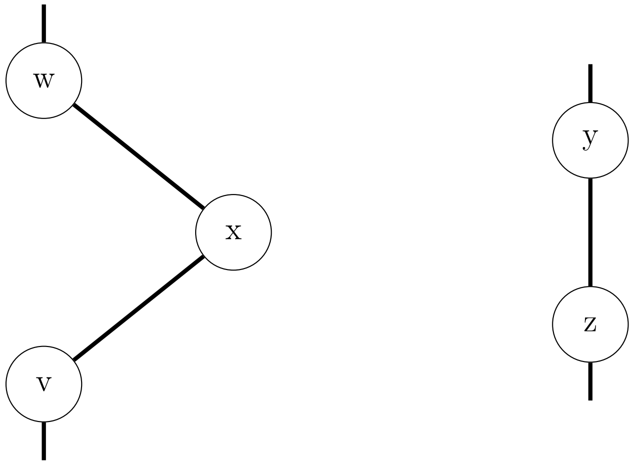

Figure 8: After moving vertex $x \in V(H_j)$ to hamiltonian path H_i.

5 Evaluation

▶ [intial cost, diff: hamiltonian-power is done after LS]

Table 5: Total cost and cable length after the local search algorithms.

	Total cost [€]	Total cable length [m]
MST approach	0	0
Hamiltonian path approach	0	0

6 Conclusion

The cabling problem asks the question of finding two spanning trees with length as well as capacity constraints onto the data and power cables. In particular, this problem has its origins from the cabling of solar tower power plants. Due to these additional constraints we are not able to solve the cabling problem in reasonable time, especially for big instances. Therefore we have designed two local search heuristic approaches that compute valid solutions within several hours for the instance PS10 that inhibits around 600 heliostats.

Besides, a partitioning problem arises due to capacity restrictions of power cables, that is, we have to install multiple power cables if the highest capacity is not sufficient to connect all heliostats. This problem is circumvented by computing a heuristic partition-solution, namely sorting angles of the solar tower and a respective heliostat and splitting the heliostat into equal-sized fields.

The first one has its basis from the minimum spanning tree, that is, the initial solution is computed by a modified version of Prim's algorithm, which takes switch installation costs into account. The local search utilizes the cycle neighborhood in addition to possible down-/upgrades of the power cables which beforehand have been computed greedily by considering capacities and length restrictions.

The second approach exploits the fact that switch costs are the main cost-factor, therefore Hamiltonian paths are considered. We first construct Hamiltonian paths greedily that is afterwards improved by a local search procedure which takes a triangulation-neighborhood into account.

► [evaluation results] The Hamiltonian path achieves a total cost of xxxx ► [cost] €compared

In future work, one may consider multiple initial solutions, that are for example randomly generated, thereby enlarging the total search space of our procedure. Also, bigger neighborhoods could be considered in order to improve solution quality, however a heuristic search within the respective neighborhood may be necessary in order to select an improving neighboring solution.

Moreover, we have left out the copper cables completely in order to simplify the heuristics because glasfiber cables do not inhibit any length or capacity restrictrions.

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